AI in Judicial Systems: A Mini-Review (2024)

1. Graph-Based Linking of Related Cases (SIGIR 2024)

- **Problem:** Keyword search misses nuanced legal similarities.
- **Approach:** Graph neural networks (GNNs) link cases via semantic/legal similarity.
- **Findings:** Graph-based retrieval improves precedent search accuracy.
- **Impact:** Helps judges/lawyers identify relevant precedents faster.

2. Judicial Knowledge Graph Construction (ICIIT 2024)

- **Problem:** Legal documents are unstructured, limiting automated reasoning.
- **Approach:** NLP extracts entities (laws, judges, verdicts) and relations; stored in RDF/OWL.
- **Findings:** Enables semantic queries (e.g., "cases citing Article 21 judged by X").
- **Impact:** Provides explainable legal analytics and decision support.

3. AI Case Prioritization for Prosecutorial Governance (AIJ 2024)

- **Problem:** Prosecutors face overwhelming backlogs.
- **Approach:** ML models rank cases by severity, social impact, conviction likelihood.
- **Findings:** Reduces backlog by ~30%, improves resource allocation.
- **Impact:** Enhances governance efficiency, but raises fairness concerns.

Comparative Analysis

Dimension	**Graph-Based Linking**	**Judicial Knowledge Graph**	**AI Case Prioritization**
Goal	Precedent retrieval	Structure legal knowledge	Optimize workload
Techniques	GNNs	NLP + entity-relation extraction	ML ranking models
Strengths	Accuracy	Explainability	Efficiency
Risks	Citation bias	Data quality	Bias in prioritization